

ELEC5660 Project 1 Phase 2 Report

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1 Task

I implemented three trajectory generators in the simulator: Smooth Only, Minimum Jerk, and Minimum Snap. All tests use Path 1.

2 Method

- Smooth Only: quintic piecewise polynomial, enforce position/velocity/acceleration continuity.
- Minimum Jerk: equality-constrained QP, minimize integral of jerk squared.
- Minimum Snap: equality-constrained QP, minimize integral of snap squared.

3 Setup

- Path: Path 1
- Controller: default
- Time allocation: equal segment time

4 Results

Metric	Smooth	Min Jerk	Min Snap
RMSE Position (m)	0.1015	0.1011	0.1097
RMSE Velocity (m/s)	0.1477	0.1279	0.1397
RMSE Yaw (deg)	2.186	2.188	2.189
Smoothness	0.068	0.068	0.068

Table 1: Path 1 comparison.

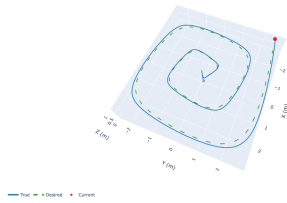


Figure 1: *
Smooth

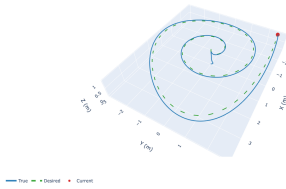


Figure 2: *
Min Jerk

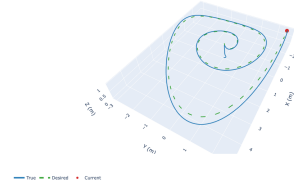


Figure 3: *
Min Snap

Figure 4: Trajectory and tracking plots on Path 1.

5 Discussion

For Path 1, Min Jerk has the best RMSE position and RMSE velocity. Min Snap is slightly worse than Min Jerk in this test. Yaw RMSE and smoothness are almost the same for all three methods.

6 Conclusion

All three methods work and pass waypoints. For this specific setup (Path 1), Minimum Jerk gives the best overall tracking result.

7 References

Course Lecture 4 slides and project starter code.